Elasticsearch Health Check and Assessment

Evidence-based evaluation of cluster health, capacity, performance, reliability, configuration, and security. No state-changing Elasticsearch operation was performed.

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Cluster	docker-cluster
Elasticsearch	8.19.20
Target	http://127.0.0.1:19200/
Overall health	High Risk | score 35 / 100
Topology	1 nodes, 3 indices, 3 shards, 59.25 KiB data
Severity counts	critical 2, high 1, medium 1, low 0, info 0

Top risks

CRITICAL **Elasticsearch HTTP traffic is not protected by TLS (http)**

Network observers can read or modify Elasticsearch traffic; credentials sent over HTTP are exposed.

CRITICAL **Credentials were sent over HTTP because plaintext authentication was explicitly allowed (http_plaintext_auth)**

The operator override transmitted secrets on an unencrypted Elasticsearch HTTP connection.

HIGH **Disk-based shard allocation thresholds are disabled (cluster.routing.allocation.disk.threshold_enabled)**

Elasticsearch can continue allocating shards to critically full nodes.

MEDIUM **Single-node cluster has no node-level redundancy (docker-cluster)**

A node or host failure makes the entire cluster unavailable.

Top resource consumers

Node by disk

Resource	Usage
1c7748d13658	69.16%

Node by JVM heap

Resource	Usage
1c7748d13658	59.00%

Node by shards

Resource	Usage
1c7748d13658	3.00 shards

Detailed findings

1. ES-SEC-001 CRITICAL
Elasticsearch HTTP traffic is not protected by TLS

Field	Value
Category	Security
Resource	transport/http
Confidence	HIGH
Threshold	HTTPS
Operational impact	Network observers can read or modify Elasticsearch traffic; credentials sent over HTTP are exposed.
Recommended action	Enable and verify TLS on the Elasticsearch HTTP interface or its trusted reverse proxy. Avoid sending credentials over HTTP.
Evidence	allow_plaintext_auth=true, credentials_used=[REDACTED], https_enabled=false

2. ES-SEC-001 CRITICAL
Credentials were sent over HTTP because plaintext authentication was explicitly allowed

Field	Value
Category	Security
Resource	transport/http_plaintext_auth
Confidence	HIGH
Threshold	HTTPS or no credentials
Operational impact	The operator override transmitted secrets on an unencrypted Elasticsearch HTTP connection.
Recommended action	Use HTTPS and keep --allow-plaintext-auth disabled except for explicitly authorized, isolated tests.
Evidence	allow_plaintext_auth=true, credentials_used=[REDACTED], https_enabled=false

3. ES-CONFIG-001 HIGH
Disk-based shard allocation thresholds are disabled

Field	Value
Category	Configuration
Resource	cluster_setting/cluster.routing.allocation.disk.threshold_enabled
Confidence	HIGH
Threshold	true
Operational impact	Elasticsearch can continue allocating shards to critically full nodes.
Recommended action	Re-enable disk thresholds through an authorized change after understanding why they were disabled.
Evidence	effective_value=false

4. ES-AVAIL-001 MEDIUM
Single-node cluster has no node-level redundancy

Field	Value
Category	Availability
Resource	cluster/docker-cluster
Confidence	HIGH
Threshold	more than one node for node-level redundancy
Operational impact	A node or host failure makes the entire cluster unavailable.
Recommended action	For production workloads, deploy additional role-appropriate nodes across independent failure domains. A single node may be intentional in development.
Evidence	number_of_nodes=1, profile=standard

Prioritized action plan

P0 Immediate

- Enable and verify TLS on the Elasticsearch HTTP interface or its trusted reverse proxy. Avoid sending credentials over HTTP.
- Use HTTPS and keep --allow-plaintext-auth disabled except for explicitly authorized, isolated tests.

P1 Urgent

- Re-enable disk thresholds through an authorized change after understanding why they were disabled.

P2 Planned

- For production workloads, deploy additional role-appropriate nodes across independent failure domains. A single node may be intentional in development.

P3 Optimization

None.

Assessment coverage and scanner telemetry

Checks	31 executed / 38 available | passed 28 skipped 7 failed 0 findings 4
API requests	11 | retried 0 | failed 0
Downloaded	86.81 KiB
Duration	515 ms

Collector	Cost	Status	HTTP	Reason
allocation_explain	MEDIUM	skipped	-	no_unassigned_shards
authenticate	LOW	success	200	-
cluster_health	LOW	success	200	-
cluster_settings	LOW	success	200	-
cluster_stats	LOW	success	200	-
data_streams	MEDIUM	skipped	-	deep_scan_disabled
ilm	HIGH	skipped	-	deep_scan_disabled
index_settings	MEDIUM	success	200	-
indices	MEDIUM	success	200	-
nodes_info	LOW	success	200	-
nodes_settings	MEDIUM	skipped	-	deep_scan_disabled
nodes_stats	MEDIUM	success	200	-
pending_tasks	LOW	success	200	-
root	LOW	success	200	-
shards	MEDIUM	success	200	-
snapshot_repositories	MEDIUM	skipped	-	deep_scan_disabled
tasks	HIGH	skipped	-	deep_scan_disabled

Methodology

This assessment combines cluster health, node resources, JVM, disk, shard and index architecture, workload pressure, lifecycle, backup, security, capacity, availability, and reliability evidence. The Elasticsearch cluster-health color is not used as the sole health decision.

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